

The Skin and Systemic Disease: A Comprehensive Study Guide

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■ Learning Objectives

By the end of this summary, you should be able to:

- ■ Recognize clues in the history and physical examination that suggest an underlying systemic disease presenting with skin findings
- ■ Differentiate between different types of drug- and infection-related skin eruptions (toxic erythema, erythema multiforme, erythema nodosum, vasculitis)
- ■ Identify characteristic rashes that serve as cutaneous markers of specific systemic diseases (sarcoidosis, SLE, IBD, endocrinopathies)
- ■ Understand skin manifestations of GI, hepatic, and hormonal diseases (zinc deficiency, acanthosis nigricans, necrobiosis lipoidica, thyroid skin changes)
- ■ Apply a systematic approach to diagnosing and managing cutaneous presentations of systemic disease

■ Key Takeaways

> ■ ****1.**** A rash that is ****widespread, non-blanching, painful, or blistering****—especially accompanied by systemic symptoms (fever, weight loss, joint pain, shortness of breath)—should raise high suspicion for systemic disease.

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> ■ ****2.**** Erythema nodosum is the ****most common panniculitis****; bilateral, painful, on the shins, resolves in 6 weeks; top causes: ****idiopathic (50%), streptococcal infection, sarcoidosis, IBD, TB, Behçet's****.

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> ■ ****3.**** Erythema multiforme is triggered most commonly by ****HSV****; systemic symptoms precede the rash by ****2–14 days****; hallmark ****target lesion**** has ****3 zones of color****.

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> ■ ****4.**** Non-blanching palpable purpuric lesions that are painful and blistering represent active ****vasculitis****—investigate for Hepatitis B/C, SLE, lymphoma, or leukemia.

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> ■ ****5.**** The skin is a ****mirror of internal disease****: acanthosis nigricans → insulin resistance/gastric cancer; necrobiosis lipoidica → diabetes; lupus pernio → sarcoidosis; pyoderma gangrenosum → IBD/RA/malignancy.

■ Main Content

■ Section 1: Clues That a Rash May Indicate Systemic Disease

When encountering a skin rash, certain features should prompt investigation for an underlying systemic condition:

- ****Rash associated with systemic symptoms:****
 - Joint pain, fever, weight loss, weakness, shortness of breath, altered bowel function
 - ■ ****Key Insight:**** The combination of skin findings + constitutional symptoms is a red flag. Think beyond the skin!
- ****Rash not responding to topical treatment:****
 - Failure to improve with emollients or topical steroids suggests an internal driver
- ****Non-blanching, palpable, purplish, fixed, painful, blistering lesion:****
 - Classic description of ****vasculitis**** — inflammation and destruction of blood vessel walls
 - ■ ****Quick Fact:**** Non-blanching = blood extravasation (purpura). Palpable = vessel wall inflammation. Together, these distinguish vasculitis from capillary fragility.
- ****Unusual pigmentation or texture changes:****
 - Hyperpigmentation → Addison's, acromegaly, pregnancy
 - Hypopigmentation → hypopituitarism
 - Thickened velvety flexures → acanthosis nigricans

■ Section 2: Characteristic Rashes Indicating Systemic Disease

These are high-yield associations frequently tested:

| Rash | Associated Systemic Diseases |

|---|---|

| Erythema multiforme | HSV, Hepatitis B/C, Mycoplasma pneumonia |

| Pyoderma gangrenosum | IBD, Rheumatoid Arthritis, Hematological malignancy |

| Erythema nodosum | IBD, Streptococcal infection, TB, Sarcoidosis, Behçet's |

| Vasculitis | Hepatitis B/C, SLE, Lymphoma, Leukemia |

■ ****Key Insight:**** These rashes are ****cutaneous markers**** of internal disease. When you see them, investigate the underlying cause!

■ Section 3: Skin Reactions Associated with Infections & Drugs

■ Toxic Erythema

- ****Presentation:****
- Widespread symmetrical maculopapular erythema (morbilliform)
- Starts trunk → spreads to limbs
- Blanching (distinguishes from vasculitis)
- ****Triggers:**** Viruses, bacteria, drugs
- ****Treatment:**** Self-limiting; emollients/topical steroids if symptomatic
- ■ ****Quick Fact:**** Most common drug eruption pattern — the skin's "nonspecific alarm"

■ Erythema Multiforme (EM)

- ****Hallmark:**** Target lesions with ****3 zones**** (dusky center, pink middle, red outer)
- ****Distribution:**** Acral (hands, feet, extremities)
- ****Most common trigger:**** HSV (infection precedes rash by ****2–14 days****)
- ****Classification:****
- EM Minor — mild, skin only
- EM Major — severe, involves mucous membranes
- ****Diagnosis:**** Clinical (characteristic target lesions)

- **Treatment:** Self-limiting; treat underlying cause; common recurrence with HSV
- **Memory Aid — "EM Doll's Eye":** Think of a bull's-eye or doll's eye target — 3 concentric zones, just like the target lesions of EM.

■ Section 4: Erythema Nodosum

The **most common panniculitis** — inflammation of subcutaneous fat:

- **Appearance:**
 - Bilateral, warm, red, raised nodules
 - Most commonly on anterior shins
 - Painful on palpation
- **Natural history:** Resolves spontaneously within **6 weeks**
- **Causes (mnemonic: "I STIBS"):**
 - **I**diopathic (~50%)
 - **S**treptococcal infection
 - **T**B
 - **I**BD
 - **B**ehçet's
 - **S**arcoidosis
- **Treatment:**
 - Treat underlying cause
 - Rest, leg elevation
 - NSAIDs for pain
 - Systemic steroids if severe
- **Key Insight:** Erythema nodosum is a **reaction pattern**, not a specific disease. Always search for the underlying trigger.

■ Section 5: Vasculitis

Inflammation and destruction of blood vessel walls:

- **Presentation:**
 - Non-blanching palpable purpura

- Painful, may blister or ulcerate
- Systemic symptoms (fever, arthralgia) may coexist
- **Key associations to investigate:**
- **Hepatitis B** → Polyarteritis nodosa (PAN)
- **Hepatitis C** → Cryoglobulinemic vasculitis
- **SLE** → Immune complex vasculitis
- **Lymphoma/Leukemia** → Paraneoplastic vasculitis
- **Pathophysiology:** Inflammation → vessel wall destruction → RBC extravasation → palpable purpura
- **Diagnostic workup:** FBC, LFTs, viral serology (HBV/HCV), ANA, cryoglobulins, complement levels, blood cultures
- **Treatment:**
- Identify and treat the underlying trigger
- Immunosuppression if progressive (corticosteroids, cyclophosphamide, rituximab)
- Supportive wound care for ulcerated lesions
- **Quick Fact:** The **size of the affected vessel** determines the clinical appearance:
- Small vessel → palpable purpura, petechiae
- Medium vessel → nodules, livedo reticularis, ulcers
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■ Section 6: Systemic Diseases with Characteristic Cutaneous Manifestations

■ Sarcoidosis

- **Lupus pernio:** Purple-violaceous infiltrative plaques on nose, cheeks, ears
- ■ **Key Insight:** Lupus pernio is highly specific for chronic systemic sarcoidosis — think of the "butterfly" counterpart for sarcoidosis just as the malar rash is for SLE.
- Other skin findings: Papules, plaques, nodules, scar infiltrates
- Associated: Bilateral hilar lymphadenopathy, uveitis, polyarthralgia

■ Systemic Lupus Erythematosus (SLE)

- **Malar (butterfly) rash:** Fixed erythema over cheeks and nasal bridge, spares nasolabial folds
- **Discoid lupus:** Scaly, atrophic plaques with scarring — may occur without systemic disease
- **Subacute cutaneous lupus:** Annular, polycyclic, non-scarring photosensitive rash

- Associated serology: ANA, anti-dsDNA, anti-Smith, low complements

■ Inflammatory Bowel Disease (IBD)

- **Pyoderma gangrenosum:** Rapidly expanding ulcer with violaceous undermined borders
- **Erythema nodosum:** Bilateral shin nodules (as discussed above)
- **Oral Crohn's:** Cobblestoning, mucosal tags, aphthous ulcers
- ■ **Key Insight:** Pyoderma can follow surgery at the site of a stoma (pathergy phenomenon)
— this is a pathognomonic clue.

■ Diabetes Mellitus

- **Necrobiosis lipoidica:** Yellow-brown atrophic plaques on shins with raised violaceous borders; waxy telangiectasia surface
- **Acanthosis nigricans:** Velvety hyperpigmentation in flexures (axillae, groin, neck)
- Other: Diabetic dermopathy ("shin spots"), glycated skin (carpal stiffness)

■ Section 7: Endocrine & Metabolic Skin Manifestations

| Condition | Skin Manifestation |

|---|---|

| **Insulin resistance / Type 2 DM** | Acanthosis nigricans |

| **Gastric adenocarcinoma** | Sudden-onset acanthosis nigricans (paraneoplastic) |

| **Diabetes mellitus** | Necrobiosis lipoidica |

| **Hypothyroidism** | Dry coarse skin, brittle hair, myxedema, loss of lateral eyebrows |

| **Hyperthyroidism** | Warm moist skin, pretibial myxedema (Graves'), onycholysis |

| **Addison's disease** | Hyperpigmentation (skin creases, buccal mucosa, scars) |

| **Acromegaly** | Coarse facial features, thickened skin, hyperhidrosis |

| **Hypopituitarism** | Pale skin (loss of MSH) |

■ **Key Insight:** Acanthosis nigricans appearing **suddenly in an adult** with weight loss should raise suspicion for **internal malignancy** (especially gastric adenocarcinoma). This is a classic paraneoplastic sign.

■ Section 8: Gastrointestinal & Hepatic Skin Manifestations

■ Zinc Deficiency (Acrodermatitis Enteropathica)

- **Classic triad:**

1. Acral and periorificial dermatitis
2. Alopecia
3. Diarrhea

- **Causes:** Malabsorption (Crohn's, celiac, short bowel syndrome), alcoholism, parenteral nutrition without zinc supplementation

- **Treatment:** Oral zinc supplementation — rapid and dramatic response

■ Hepatic Disease

- **Hepatitis B/C:** Vasculitis (PAN with HBV; cryoglobulinemia with HCV)

- **Chronic liver disease:** Spider naevi, palmar erythema, leukonychia, jaundice, pruritus

- **Primary biliary cholangitis:** Pruritus, xanthomata, pigmentation

■ **Quick Fact:** **Spider naevi** = central arteriole with radiating capillary "legs"; pressure on the center causes blanching of the entire lesion. They are associated with hyperestrogenism in liver disease.

■ Connections & Applications

Understanding how skin manifestations connect to systemic disease is critical for clinical practice:

- **The Skin as a Diagnostic Window:**

- Many systemic diseases have pathognomonic skin findings that appear **before** other organ involvement

- A careful skin exam can lead to early diagnosis of internal malignancy, autoimmune disease, or infection

- **Cross-Disciplinary Relevance:**

- **Gastroenterology:** IBD → erythema nodosum, pyoderma gangrenosum

- **Endocrinology:** Diabetes → necrobiosis lipoidica, acanthosis nigricans

- **Rheumatology:** SLE → malar rash, discoid lupus

- **Pulmonology:** Sarcoidosis → lupus pernio

- **Oncology:** Acanthosis nigricans → gastric adenocarcinoma

- **Infectious Disease:** EM → HSV reactivation; EN → TB

- **Clinical Decision Pathway:**

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Rash + Systemic Symptoms → Is it blanching or non-blanching?

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Blanching → Think toxic erythema, EM

Non-blanching → Think vasculitis → Investigate cause (HBV, HCV, SLE, malignancy)

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Nodular → Think EN → Investigate (IBD, sarcoidosis, streptococcal, TB)

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Ulcer → Think pyoderma gangrenosum → Investigate (IBD, RA, malignancy)

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■ Quick Review Checklist

- [x] Can you identify the features of a rash that suggest systemic disease?
- [x] Do you know the association between EM and HSV, and the 2–14 day prodrome?
- [x] Can you recall the 6 most common causes of erythema nodosum?
- [x] Do you understand why non-blanching palpable purpura indicates vasculitis?
- [x] Can you list the skin manifestations of sarcoidosis, SLE, IBD, and diabetes?
- [x] Do you know that sudden acanthosis nigricans in an adult suggests gastric cancer?
- [x] Can you describe the classic triad of zinc deficiency?
- [x] Are you familiar with thyroid-related skin changes (hypo- vs. hyperthyroidism)?
- [x] Do you know the clinical difference between small-vessel and medium-vessel vasculitis?
- [x] Can you apply the systematic approach to evaluate a rash for systemic disease?

■ Memory Aids & Mnemonics

| Mnemonic | Condition | Details |

|---|---|---|

| **"I STIBS"** | Erythema nodosum causes | **I**diopathic, **S**treptococcal, **TB**, **IBD**, **B**ehçet's, **S**arcoidosis |

| **"EM Doll's Eye"** | Erythema multiforme | Target lesion = 3 concentric zones like a bull's-eye |

| **"ABCDE of Paraneoplastic AN"** | Acanthosis nigricans | **A**brupt, **B**ody cavity cancer, **C**heck for **D**iabetes & **E**ndocrine disease |

| **"BLT with Mayo"** | Vasculitis associations | **"B"** (Hep B → PAN), **L**ow complements, **T**umor. Think mayo — creamy = cryoglobulins." |

| **"ZINC Triad"** | Zinc deficiency | **Z**inc dermatitis (acral/periorificial) → **I**nfection resistance drop → **N**ails/hair affected → **C**hronic diarrhea |

| **"Butterfly vs Lupus Pernio"** | SLE vs Sarcoidosis | SLE = butterfly rash; Sarcoidosis = lupus pernio (purple plaques on nose) |

| **"Warm vs Cold Skin"** | Thyroid skin changes | **Warm** = Hyperthyroidism; **Cold/Dry** = Hypothyroidism |

■ **Analogy — The Skin as a "Weather Vane":**

Just as a weather vane shows which direction the wind is blowing, the skin shows the direction of **internal disease**. You don't treat the weather vane — you respond to what it's telling you. A rash is the body's way of pointing to the real problem inside.

■ This study guide is designed for medical students and clinicians. Always correlate skin findings with clinical context, history, and investigations for accurate diagnosis and management.